

Total No. of Questions : 10]

SEAT No. :

PA-348

[Total No. of Pages : 2

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B.E. (Mechanical)

PRODUCT DESIGN AND DEVELOPMENT

(2015 Pattern) (Semester - II) (Elective - IV) (402050 C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of calculators is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Explain phases of product design. **[5]**

b) Differentiate between product development and product design. **[5]**

OR

Q2) a) Explain customer need with suitable example. **[5]**

b) Explain the concept of Technology forecasting. **[5]**

Q3) a) Explain morphological analysis with suitable example. **[5]**

b) How standardization, simplification and specialization are related to product design'? **[5]**

OR

Q4) a) Discuss product testing in detail. **[5]**

b) Explain the process of concept embodiment. **[5]**

P.T.O.

- Q5)** a) What is product tear down? Explain the tear down process in detail. [8]
b) Explain force flow diagram with suitable example. [8]

OR

- Q6)** a) Explain any two types of product architecture. [8]
b) Explain following tools in benchmarking [8]
i) Intended Assembly Cost Analysis.
ii) Trend Analysis.

- Q7)** a) Explain guideline for design for manufacturing. [8]
b) Discuss guideline for design for environment. [8]

OR

- Q8)** a) Discuss guidelines design for Reliability. [8]
b) What are guidelines for design for safety? Discuss and elaborate. [8]

- Q9)** a) What is Life cycle management? Discuss the applications of PLM. [10]
b) Explain the components of PDM in detail. [8]

OR

- Q10)** a) Discuss the concept of product data and product workflow in PLM. [10]
b) Explain Five phases of PLC in detail with example. [8]

